1. HOLLOW SQUARE PATTERN WITH X PATTERN S6

Hollow Square Pattern with X pattern inside(Made of 0's and 1's) using functions

Input Format

INT

Constraints

NULL

Output Format

MATRIX

Sample Input 0

5

Sample Output 0

11111

11011

10101

11011

11111

Explanation 0

Box and X shape = 1

Sample Input 1

7

Sample Output 1

1111111

1100011

1010101

1001001

1010101

1100011

1111111

```
import java.io.*;
import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int a=sc.nextInt();
        int i,j;
        for(i=1;i<=a;i++){
            for(j=1;j<=a;j++){
                if(i==1 || i==a || j==1 || j==a || j==i || (a+1==j+i)){
                    System.out.print("1");
                }else{
                    System.out.print("0");
                }
            }
            System.out.println();
        }
    }
}
```

2. HOLLOW TRIANGLE STAR PATTERN

Hollow triangle star pattern in C (nested looping)

Input Format

INT

Constraints

NULL

Output Format

Pattern : Hollow triangle star (based on the given number)

Sample Input 0

5

Sample Output 0

```
*
**
* *
*  *
*****
```

```
import java.io.*;
import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int a=n/2+1;
        int i,j;
        for(i=1;i<=n;i++){
            for(j=1;j<=n;j++){
                if(j==1 || i==n || j==i){
                System.out.print("*");
            }else{
                System.out.print(" ");
            }
        }
        System.out.println();
        }
    }
}
```

3. HOLLOW PYRAMID STAR PATTERN 2

Print Hollow Pyramid Star Pattern

Input Format

int

Constraints

NULL

Output Format

pattern shape

Sample Input 0

5

Sample Output 0

```
    *
  *   *
 *     *
*       *
*********
```

Sample Input 1

9

Sample Output 1

```
     *
    * *
   *    *
  *      *
 *         *
*           *
*             *
*               *
*****************
```

```
import java.io.*;
import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int a=sc.nextInt();
        int b=a*2-1;
        for(int i=1;i<=a;i++){
            for(int j=1;j<=b;j++){
                if(i==a || (j<=a && a+1==j+i) || (j>a && a-1==j-i)){
                    System.out.print("*");
                }else{
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

4. NUMBER WITH 1 DIAGONAL PATTERN

Pattern n = 9 98765432* 9876543*1 987654*21 98765*321 9876*4321 987*54321 98*654321 9*7654321 *87654321

Input Format

int

Constraints

NULL

Output Format

Pattern

Sample Input 0

9

Sample Output 0

98765432*

9876543*1

987654*21

98765*321

9876*4321

987*54321

98*654321

9*7654321

*87654321

Sample Input 1

8

Sample Output 1

8765432*

876543*1

87654*21

8765*321

876*4321

87*54321

8*654321

*7654321

```
import java.io.*;
import java.util.*;
public class Solution {
    public static void main(String[] args) {
        /* Enter your code here. Read input from STDIN. Print output to STDOUT.
Your class should be named Solution. */
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int i,j;
        for(i=1;i<=n;i++){
            for(j=n;j>=1;j--){
                if(i==j){
                System.out.print("*");
            }else{
                System.out.print(j);
            }
        }
        System.out.println();
        }
    }
}
```

5. RECTANGLE STAR PATTERN 1

Print Rectangle Star Pattern length and width

Input Format

INT INT

Constraints

NULL

Output Format

Shape

Sample Input 0

4 6

Sample Output 0

******

******

******

******

Sample Input 1

6 4

Sample Output 1

****

****

****

****

****

****

```
import java.io.*;
import java.util.*;
public class Solution {
   public static void main(String[] args) {
      Scanner sc=new Scanner(System.in);
      int n=sc.nextInt();
      int a=sc.nextInt();
      int i,j;
      for(i=1;i<=n;i++){
         for(j=1;j<=a;j++){
            System.out.print("*");
      }
      System.out.println();
      }
   }
}
```

6. ZIGZAG PATTERN NUMBER

ZigZag Pattern Number

Input Format

INT

Constraints

NULL

Output Format

Pattern

Sample Input 0

4

Sample Output 0

001

002 007

003 006 008

004 005 009 010

Sample Input 1

10

Sample Output 1

001

002 019

003 018 020

004 017 021 034

005 016 022 033 035

006 015 023 032 036 045

007 014 024 031 037 044 046

008 013 025 030 038 043 047 052

009 012 026 029 039 042 048 051 053

010 011 027 028 040 041 049 050 054 055

Sample Input 2

19

Sample Output 2

001

002 037

003 036 038

004 035 039 070

005 034 040 069 071

006 033 041 068 072 099

007 032 042 067 073 098 100

008 031 043 066 074 097 101 124

009 030 044 065 075 096 102 123 125

010 029 045 064 076 095 103 122 126 145

011 028 046 063 077 094 104 121 127 144 146

012 027 047 062 078 093 105 120 128 143 147 162

013 026 048 061 079 092 106 119 129 142 148 161 163

014 025 049 060 080 091 107 118 130 141 149 160 164 175

015 024 050 059 081 090 108 117 131 140 150 159 165 174 176

016 023 051 058 082 089 109 116 132 139 151 158 166 173 177 184

017 022 052 057 083 088 110 115 133 138 152 157 167 172 178 183 185

018 021 053 056 084 087 111 114 134 137 153 156 168 171 179 182 186 189

019 020 054 055 085 086 112 113 135 136 154 155 169 170 180 181 187 188 190

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;
public class Solution {
   public static void main(String[] args) {
      Scanner sc=new Scanner(System.in);
      int n=sc.nextInt();
      int arr[][]=new int[n][n];
      int i,j,num=1;
      for(i=0;i<n;i++){
         if(i%2==0){
            for(j=i;j<n;j++){
               arr[j][i]=num;
               num++;
            }
         }else{
            for(j=n-1;j>=i;j--){
               arr[j][i]=num;
               num++;
            }
         }
      }
      for(i=0;i<n;i++){
         for(j=0;j<=i;j++){
            System.out.printf("%03d ",arr[i][j]);
         }
         System.out.println();
      }
   }
}
```

7. NUMBER TRIANGLE PATTERN S

Number triangle pattern n = 9

0 909 89098 7890987 678909876 56789098765 4567890987654 345678909876543 23456789098765432 1234567890987654321

Input Format

INT

Constraints

NULL

Output Format

pattern

Sample Input 0

4

Sample Output 0

0

404

34043

2340432

123404321

Sample Input 1

5

Sample Output 1

0

505

45054

3450543

234505432

12345054321

```
import java.io.*;
import java.util.*;
public class Solution {
   public static void main(String[] args) {
      Scanner sc=new Scanner(System.in);
      int n=sc.nextInt();
      for(int i=0;i<=n;i++){
         for(int j=n-i+1;j<=n;j++){
               System.out.print(j);
            }
         /*if(i>0){
            System.out.print(0);
         }*/
         System.out.print("0");
         for(int j=n;j>=n-i+1;j--){
            System.out.print(j);
         }
         System.out.println();
      }
   }
}
```